

## Gradient-intercept form



1 Consider the equation of a line  $y = mx + c$ .

a What does  $m$  represent?

b What does  $c$  represent?

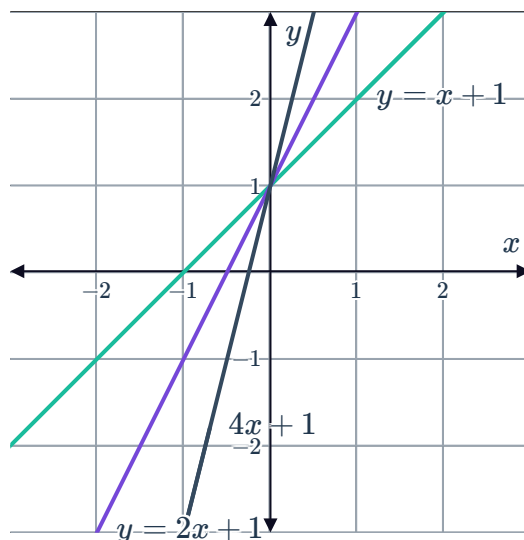
2 Consider the following three linear equations with their graphs plotted on a number plane:

- Equation 1:  $y = x + 1$
- Equation 2:  $y = 2x + 1$
- Equation 3:  $y = 4x + 1$

a What do all of the equations have in common?

b What do all of the graphs have in common?

c Describe all lines that have the form:  
 $y = mx + 1$



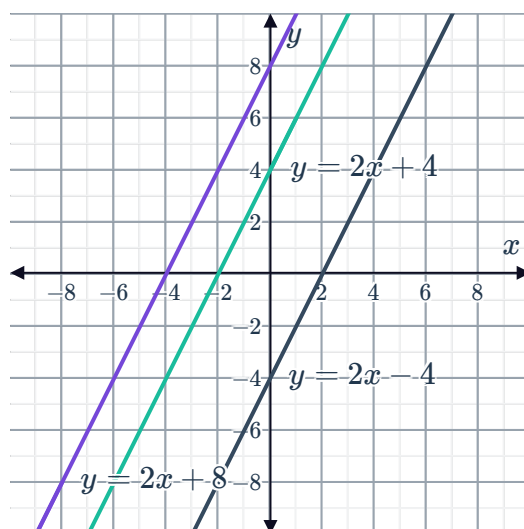
3 Consider the following three linear equations with their graphs plotted on a number plane:

- Equation 1:  $y = 2x + 4$
- Equation 2:  $y = 2x + 8$
- Equation 3:  $y = 2x - 4$

a What do all of the equations have in common?

b What do all of the graphs have in common?

c Describe all lines that have the form:  
 $y = 2x + c$



4 Determine whether or not the gradients of the following pairs of equations are equal:

a i  $y = 7x$  ii  $y = 7x - 4$

b i  $y = 7x + 2$  ii  $y = \frac{x}{7} + 4$

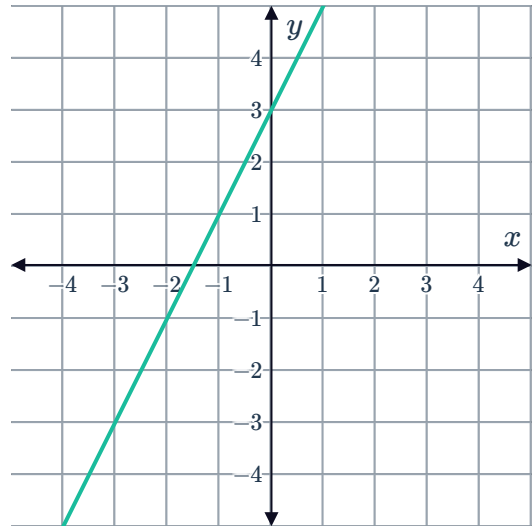
c i  $y = 6 - 7x$  ii  $y = 6x + 7$

d i  $y = 4 + 7x$  ii  $y = 7x - 10$

5 Consider the line plotted below:

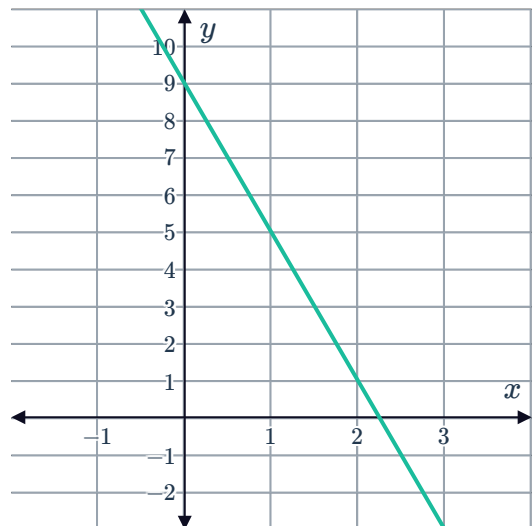


- a State the the  $y$ -intercept.
- b State the gradient.



6 Consider the line plotted below:

- a By how much does the  $y$ -value change as the  $x$ -value increases by 1?
- b State the gradient of the line.



7 State whether or not the following lines have a  $y$ -intercept:

- |                |             |                 |                 |
|----------------|-------------|-----------------|-----------------|
| a $y = 2$      | b $5x = 4y$ | c $y = -4x + 1$ | d $x = 1$       |
| e $y = 6x - 2$ | f $y = 3x$  | g $2x + 6y = 3$ | h $y + 4 = -5x$ |

8 Consider the following linear equations:

- i State the value of the gradient,  $m$ .
- ii State the  $y$ -intercept,  $c$ .

- |                   |                   |                   |                           |
|-------------------|-------------------|-------------------|---------------------------|
| a $y = -2x + 9$   | b $y = -4x - 8$   | c $y = 8x + 6$    | d $y = -1 - \frac{9x}{2}$ |
| e $3y = 12x - 15$ | f $-9x + 9y = 27$ | g $3x - 10y = -2$ | h $16x + 12y = 10$        |

9 Sketch the following lines on a number plane



- a The line with a  $y$ -intercept of  $-2$  and gradient of  $-3$ .
- b The line with a  $y$ -intercept of  $3$  and gradient of  $-\frac{3}{2}$ .
- c The line  $y = 2x + 5$ .
- d The line  $y = \frac{1}{2}x - 1$ .

10 For each of the following equations:

- i Find the  $y$ -value of the  $y$ -intercept of the line.
- ii Find the  $x$ -value of the  $x$ -intercept of the line.
- iii Find the value of  $y$  when  $x = 3$ .
- iv Sketch the line on a number plane.

a  $y = -2x$

b  $y = \frac{2x}{3} - 4$

11 For each of the following lines:

- i Find the  $y$ -coordinate of the  $y$ -intercept of the line.
- ii Hence, write the equation of the line in gradient-intercept form.
- iii Find the  $x$ -coordinate of the  $x$ -intercept of the line.
- iv Sketch the line on a number plane.

a A line has gradient  $\frac{4}{5}$  and passes through the point  $(-10, 4)$ .

b A line has gradient  $-2$  and passes through the point  $(3, -8)$ .

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## Equation of a line

12 A line has a gradient of  $-3$  and intercepts the  $y$ -axis at  $-2$ .

- a Find the equation of the line in the form  $y = mx + c$ .
- b State whether the point  $(-2, 4)$  lies on this line.

13 A line has a gradient of  $-3$  and cuts the  $y$ -axis at  $8$ .

- a Find the equation of the line in the form  $y = mx + c$ .
- b State whether the point  $(8, -31)$  lies on this line.

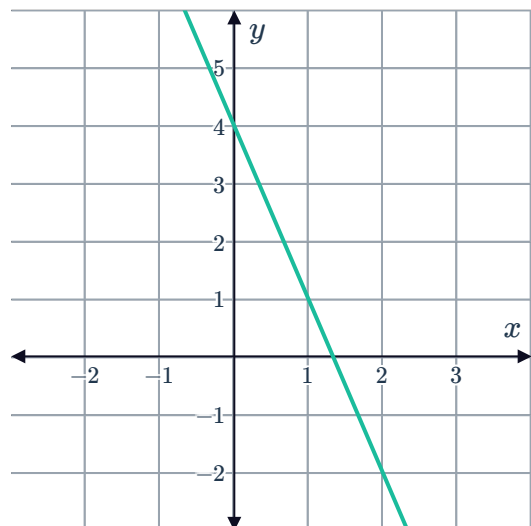
14 Find the equations of the following in the form  $y = mx + c$ :



- a A line that has the same gradient as  $y = 9 - 8x$  and the same  $y$ -intercept as  $y = -5x - 3$ .
- b A line whose gradient is 2 and crosses the  $y$ -axis at 5.
- c A line whose gradient is  $-6$  and crosses the  $y$ -axis at 9.
- d A line whose gradient is  $-8$  and crosses the  $y$ -axis at  $-9$ .
- e A line whose gradient is  $-\frac{3}{4}$  and intercepts the  $y$ -axis at 3.
- f A line whose gradient is  $\frac{4}{3}$  and goes through the point  $(0, 3)$ .
- g A line that has a gradient of  $-2$  and passes through  $(-6, -3)$
- h A line whose gradient is 8 and goes through the point  $(0, -4)$ .
- i A line whose gradient is 0 and goes through the point  $\left(0, \frac{2}{13}\right)$ .

15 Consider the line plotted on the number plane.

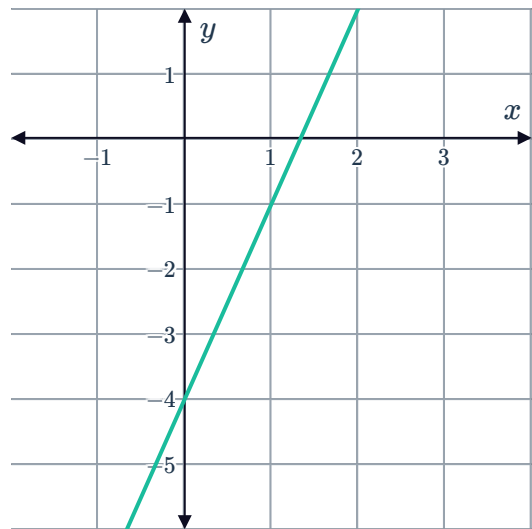
- a State the values of:
  - i The gradient,  $m$ .
  - ii The  $y$ -intercept,  $c$ .
- b Find the equation of the line in gradient-intercept form.
- c Find the value of  $y$  when  $x = 27$ .



16 Consider the line plotted on the number plane

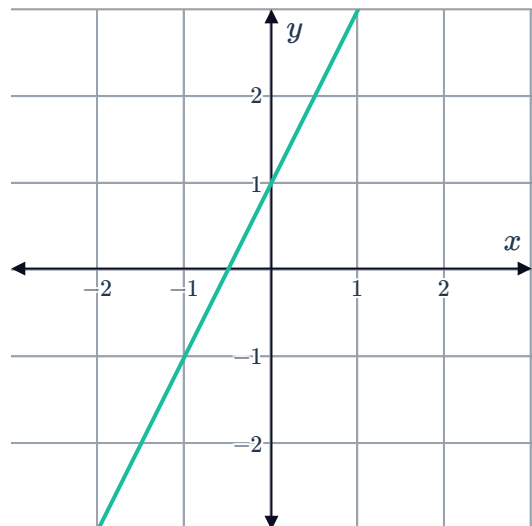


- a Find the equation of the line in gradient-intercept form.
- b Find the value of  $y$  when  $x = 50$ .



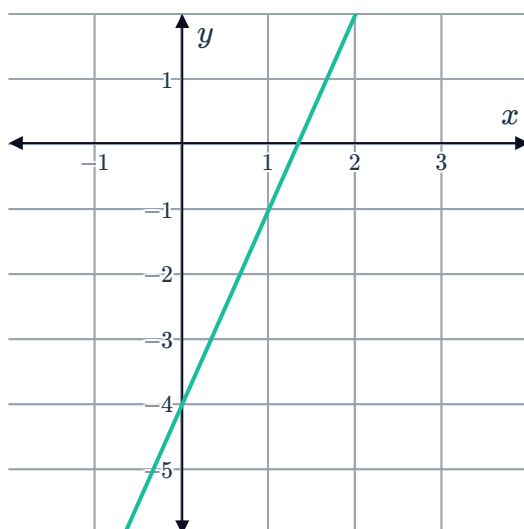
17 Consider the line plotted on the number plane.

- a Find the equation of the line in gradient-intercept form.
- b Find the value of  $y$  when  $x = 29$ .

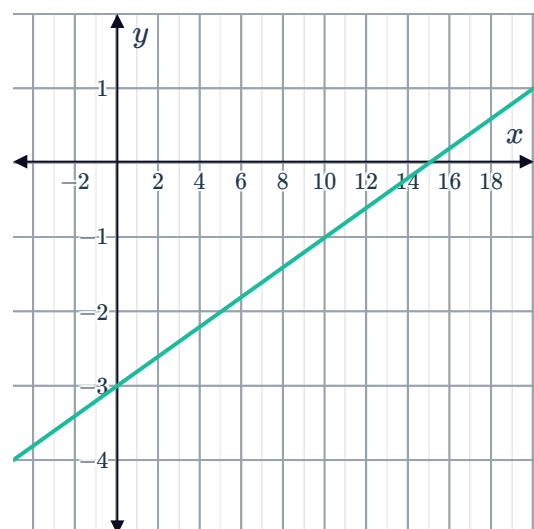


18 Find the equations of the following lines in gradient-intercept form:

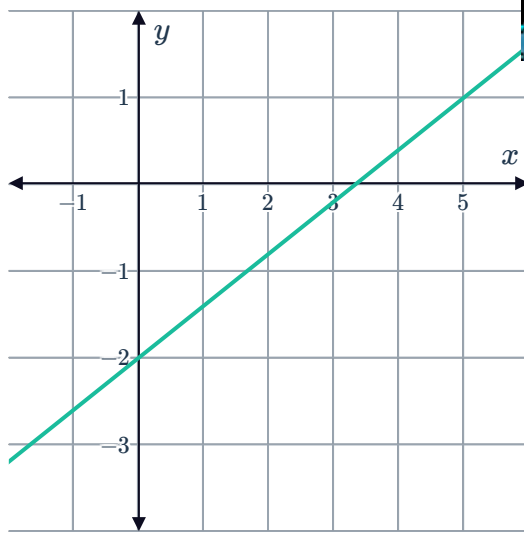
a



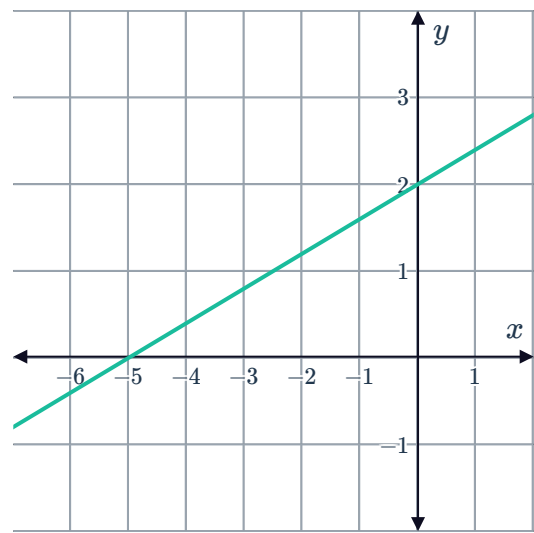
b



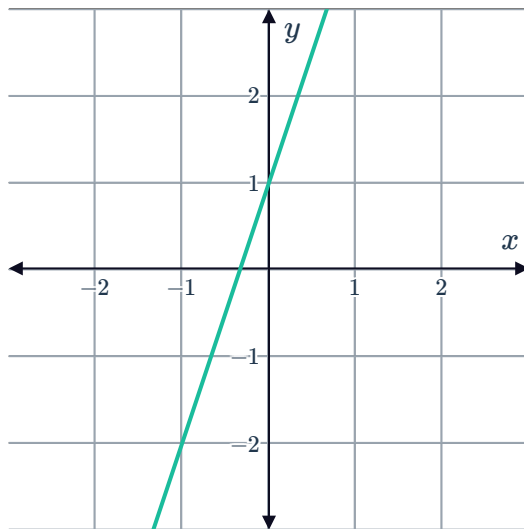
c



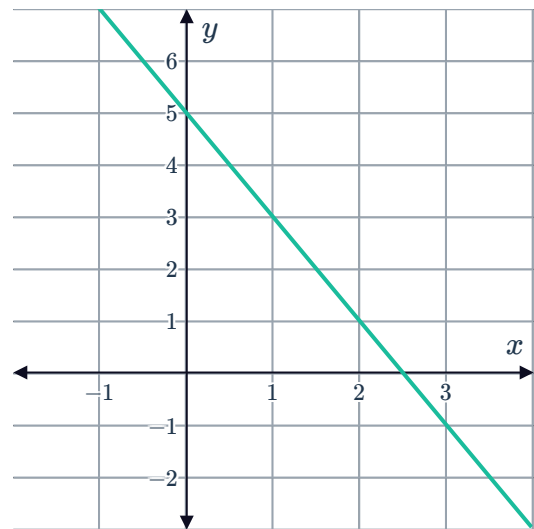
d



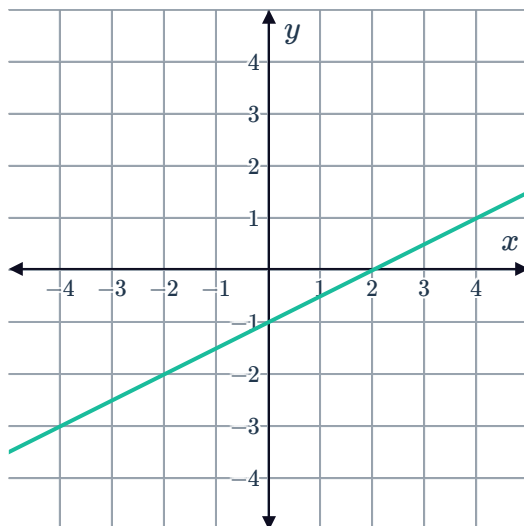
e



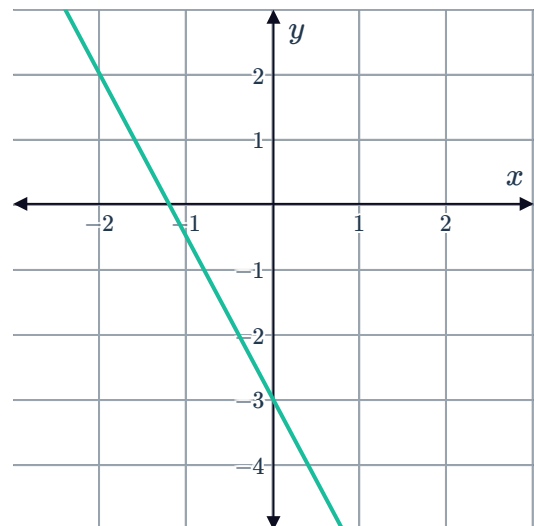
f



g



h



**19** For the following lines passing through the  two points:

- i** Hence, state the gradient of the line.
- ii** Find the equation of the line in the form  $y = mx + c$ .

- a**  $(0, 2)$  and  $(2, 6)$
- b**  $(0, -9)$  and  $(5, 1)$
- c**  $(0, 2)$  and  $(-7, 44)$
- d**  $(0, -3)$  and  $(1, 4)$

**20** For each of the following linear equations:

- i** Rewrite it in the form  $y = mx + c$ .
- ii** State the gradient of the line,  $m$ .
- iii** State the  $y$ -intercept of the line,  $c$ .

- a**  $y = \frac{-4x + 16}{4}$
- b**  $9x - y - 8 = 0$ .
- c**  $y = 3(4x - 3)$
- d**  $y = 6(3x - 2)$
- e**  $3x - 9y - 27 = 0$ .
- f**  $3x - 4y - 28 = 0$

**21** Consider the lines with the following equations:

- Line A:  $5x + 3y + 5 = 0$
- Line B:  $7x + 6y - 3 = 0$

- a** Express the lines in the form  $y = mx + c$ .
- b** State which line is steeper, A or B.

**22** Determine which of the following lines are steeper:  $2x + 5y - 5 = 0$  or  $4x + 4y + 1 = 0$ .

**23** A straight line has gradient  $-1$  and goes through the points  $(0, 2)$  and  $(a, -6)$ .

- a** Write the equation of the line in the form  $y = mx + b$ .
- b** Find the value of  $a$ .



## Gradients of parallel lines

**24** Determine whether the following statements about two parallel lines are true or false.

- a** The  $y$ -value is changing at the same rate on both lines.
- b** They intersect at one point.
- c** They have the same value of  $c$  in  $y = mx + c$ .
- d** They have the same value of  $m$  in  $y = mx + c$ .
- e** They are equidistant from each other.

**25** State whether the given pairs of lines are parallel:

- a**  $y = -2x - 5$  and  $y = -2x - 8$
- b**  $y = 7x + 8$  and  $y = -5x + 8$
- c**  $y = -3x - 2$  and  $y = -3x + 9$
- d**  $y = -6x - 5$  and  $y = -6x$

**26** Find the gradient of the following lines:

- a** A line parallel to a line with gradient  $-2$ .
- b** Any line that is parallel to  $y = -7 + 4x$ .

**27** State whether the following lines are parallel to  $y = 7x + 3$ .

- a**  $y = 7x - 3$
- b**  $y = 6x + 3$
- c**  $y = 7x$
- d**  $y = -7x + 3$

**28** State whether the following lines are parallel to  $y = -3x + 2$ .

- a**  $y = 3x$
- b**  $y = -\frac{2x}{3} + 8$
- c**  $-3y - x = 5$
- d**  $y = -10 - 3x$





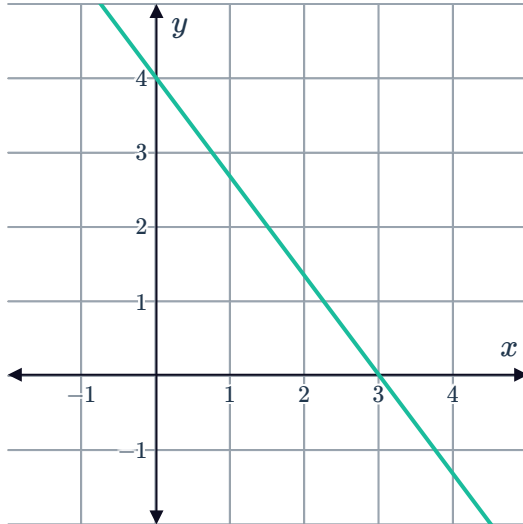
## Point-gradient formula

29 For the following graphs:

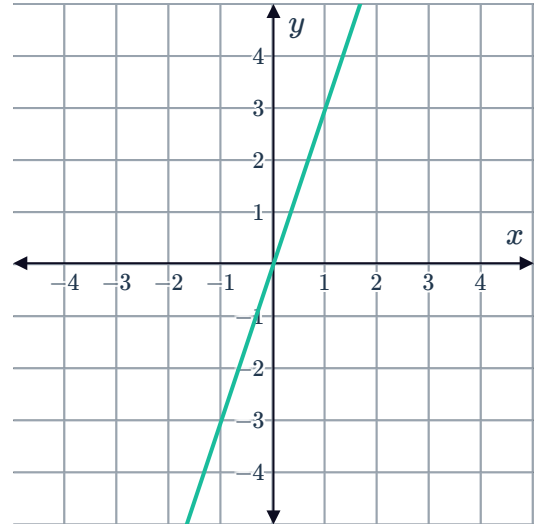
i State the value of the  $x$ -intercept.

ii State the value of the  $y$ -intercept.

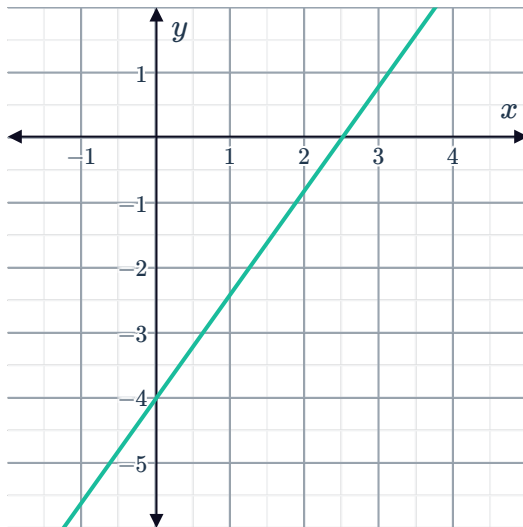
a



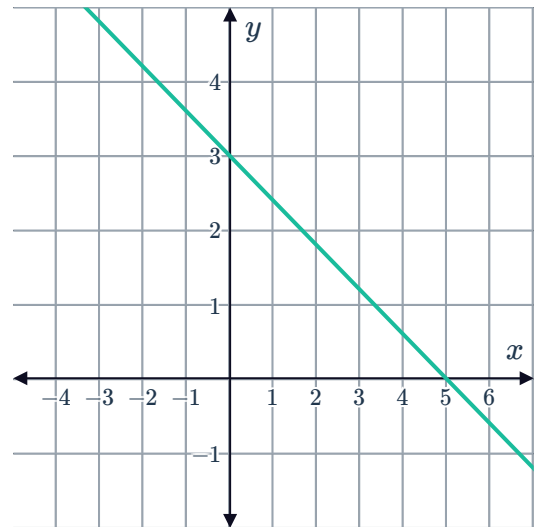
b



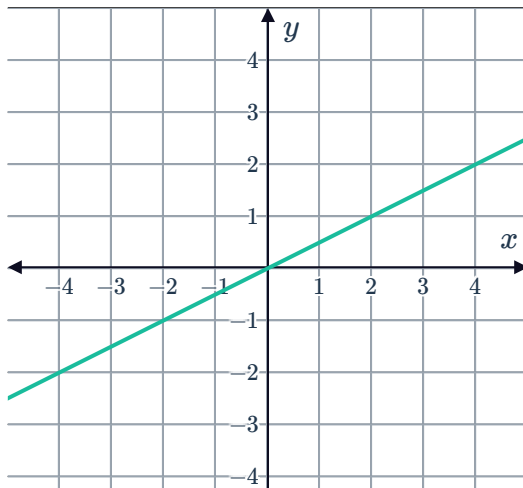
c



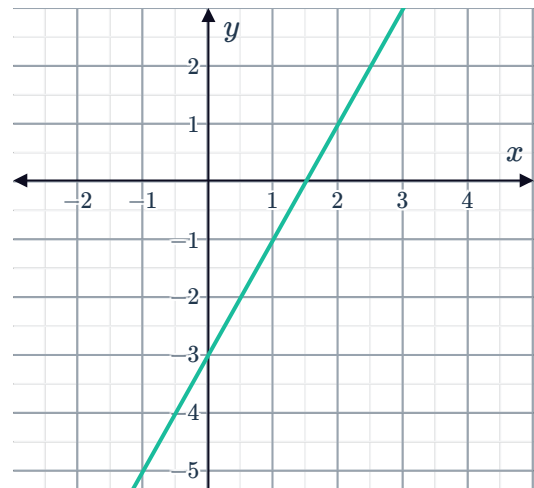
d



e



f





**30** For each of the following tables of values:

- i Find the gradient,  $m$ .
- ii Find the  $y$ -intercept,  $c$ .
- iii Write the equation of the line expressing the relationship between  $x$  and  $y$ .
- iv Complete the table of values.

**a**

|     |   |   |   |    |    |    |
|-----|---|---|---|----|----|----|
| $x$ | 0 | 1 | 2 | 3  | 4  | 24 |
| $y$ | 0 | 4 | 8 | 12 | 16 |    |

**b**

|     |   |    |    |    |    |    |
|-----|---|----|----|----|----|----|
| $x$ | 0 | 1  | 2  | 3  | 4  | 21 |
| $y$ | 9 | 14 | 19 | 24 | 29 |    |

**c**

|     |     |     |     |     |     |    |
|-----|-----|-----|-----|-----|-----|----|
| $x$ | 0   | 1   | 2   | 3   | 4   | 25 |
| $y$ | -23 | -21 | -19 | -17 | -15 |    |

**d**

|     |    |    |    |    |   |    |
|-----|----|----|----|----|---|----|
| $x$ | 0  | 1  | 2  | 3  | 4 | 70 |
| $y$ | 27 | 22 | 17 | 12 | 7 |    |

31 For each of the following tables of values:



- i Find the equation of the line expressing the relationship between  $x$  and  $y$ .
- ii Complete the table of values.

a

|     |   |    |    |    |    |
|-----|---|----|----|----|----|
| $x$ | 1 | 2  | 3  | 4  | 19 |
| $y$ | 5 | 10 | 15 | 20 |    |

b

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | 1  | 2  | 3  | 4  | 16 |
| $y$ | -3 | -6 | -9 | -6 |    |

c

|     |   |   |   |    |    |
|-----|---|---|---|----|----|
| $x$ | 1 | 2 | 3 | 4  | 60 |
| $y$ | 1 | 4 | 7 | 10 |    |

d

|     |     |     |     |     |    |
|-----|-----|-----|-----|-----|----|
| $x$ | 1   | 2   | 3   | 4   | 80 |
| $y$ | -14 | -22 | -30 | -38 |    |

e

|     |    |    |    |    |     |
|-----|----|----|----|----|-----|
| $x$ | 1  | 2  | 3  | 4  | -15 |
| $y$ | 82 | 74 | 66 | 58 |     |

32 Find the equation that corresponds to each of the following tables:

a

|     |    |   |   |    |    |
|-----|----|---|---|----|----|
| $x$ | -1 | 0 | 1 | 2  | 3  |
| $y$ | 7  | 4 | 1 | -2 | -5 |

b

|     |   |   |    |    |    |
|-----|---|---|----|----|----|
| $x$ | 1 | 2 | 3  | 4  | 5  |
| $y$ | 5 | 8 | 11 | 14 | 17 |

c

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| $x$ | -8  | -7  | -6  | -5  | -4  |
| $y$ | -37 | -32 | -27 | -22 | -17 |

d

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | 3  | 4  | 5  | 6  | 7  |
| $y$ | -1 | -3 | -5 | -7 | -9 |

**33** Find the equation of the following lines:



- a** A line that passes through the point A  $(-5, -4)$  and has a gradient of  $-4$ .
- b** A line that passes through the point A  $\left(-\frac{4}{5}, -4\right)$  and has a gradient of  $2$ .
- c** A line that passes through Point A  $(-4, 3)$  and has a gradient of  $4$ .
- d** A line that passes through Point A  $(7, -6)$  and has a gradient of  $-3$ .
- e** A line that passes through the point A  $(3, 5)$  and has a gradient of  $-\frac{5}{2}$ .
- f** A line that passes through the point A  $(4, 3)$  and has a gradient of  $-3\frac{1}{3}$ .
- g** A line that passes through the point A  $(-4, 3)$  and has a gradient of  $-9$ .
- h** A line that passes through the point A  $\left(-\frac{5}{9}, 7\right)$  and has a gradient of  $7$ .
- i** A line that passes through the point A  $(8, 1)$ , and has a gradient of  $\frac{5}{2}$ .
- j** A line that passes through the point A  $(-4, 5)$  and has a gradient of  $3\frac{1}{2}$ .

**34** For each of the following lines:

- i** Find the equation of the line.
  - ii** Sketch the graph of the line.
- a** A line has gradient  $2$  and passes through the point  $(-5, -3)$ .
  - b** A line has gradient  $-\frac{3}{2}$  and passes through the point  $(-2, 2)$ .
  - c** A line has gradient  $-\frac{2}{5}$  and passes through the point  $(-10, 2)$ .
  - d** A line has gradient  $-3$  and passes through the point  $(2, -12)$ .

**35** Consider the line with equation  $2x + y - 8 = 0$ .

- a** Find the  $x$ -intercept of the line.
- b** Hence, find the equation of a line with a gradient of  $-4$  that passes through the  $x$ -intercept of the given line.

**36** Find, in general form, the equation of a line which has a gradient of  $\frac{4}{7}$  and cuts the  $x$ -axis at  $-10$ .

**37** For each of the following lines:



- i Find the gradient of the line.
  - ii Find the equation of the line.
- 
- a A line passes through the points  $(2, -7)$  and  $(-5, 6)$ .
  - b A line passes through the points  $(3, -3)$  and  $(5, -11)$ .
  - c A line passes through the points  $A(-6, 7)$  and  $B(-8, -4)$ .

**38** Identify which of the following equations of straight lines have a gradient of 5 and pass through the point  $A(-1, -4)$ :

- a  $\frac{y+4}{x+1} = 5$       b  $\frac{x+1}{y+4} = 5$       c  $\frac{-4-y}{-1-x} = 5$       d  $\frac{y+1}{x+4} = 5$

**39** Write down the equations of three lines that pass through the point  $(1, 3)$ . Explain how your lines are different.

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## Applications

**40** A line has a gradient of  $\frac{3}{10}$  and passes through the midpoint of  $A(-6, -6)$  and  $B(8, 8)$ .

- a Find the coordinates of  $M$ , the midpoint of  $AB$ .
- b Find the equation of the line in general form.

**41** Consider the lines  $L_1$ ,  $y = -4x + 5$ , and  $L_2$ ,  $y = x - 1$ .

- a Find the midpoint  $M$  of their  $y$ -intercepts.
- b Find the equation of the line that goes through the point  $M$  and has gradient  $\frac{1}{3}$ . Express the equation in general form.

**42** A circle with centre  $C(11, 13)$  has a diameter with end points  $A(5, 14)$  and  $B(p, q)$ .

- a Find the value of  $p$ .
- b Find the value of  $q$ .
- c Find the equation of the line passing through  $B$  with gradient  $\frac{9}{2}$ .